Credit Card Default Prediction

Using Machine Learning to Identify High-Risk Borrowers

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Project Overview

Goal: Predict the probability that a credit card client will default on their next month's payment using historical payment and financial data.

Dataset:
Default Payments of Credit Card Clients in Taiwan (2005)

30,000	25	22%
Records	Features	Default Rate

Project Workflow

1. **Data Loading & Inspection**
   Loaded dataset, reviewed data types, shape, and null values
2. **Data Cleaning**
   Renamed columns, removed 35 duplicates, standardized categoricals
3. **Exploratory Analysis**
   Visualized distributions, correlations, and feature relationships
4. **Feature Engineering**
   Created 4 new features
5. **Model Building & Evaluation**
   Trained three classifiers; evaluated using ROC-AUC, F1, and confusion matrices

Data Cleaning & Preprocessing

Column Renaming: Columns were renamed for clarity (e.g. default payment next month' → DEFAULT)

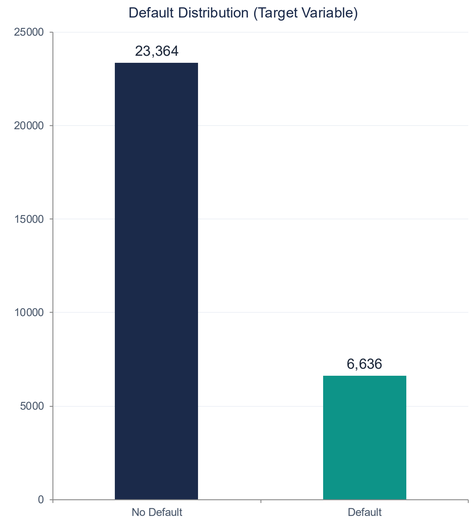
Duplicates Removed: 35 duplicate rows identified and dropped

Category Standardization: Unusual values in Education and Marital Status were grouped into an 'Other' category

Encoding: Categorical variables (Sex, Education, Marital Status) were converted into numeric form so models can process them

Feature Scaling: numerical features were normalized so that large values (e.g. credit limits) don't unfairly dominate the model

Exploratory Data Analysis (EDA)



Key Insights

Target Imbalance: ~22% of clients defaulted. Class imbalance must be explicitly addressed in modeling.

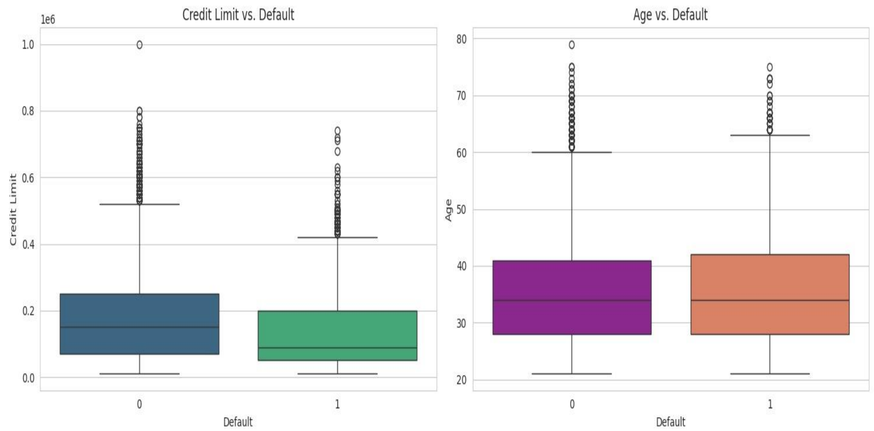
Payment Status (PAY_X):
Strongest predictor — clients with delayed or missed payments are far more likely to default.

Credit Limit (LIMIT_BAL):
Lower credit limits are associated with higher default rates.

Bill & Payment Amounts: High inter-correlation among monthly amounts; engineered features help summarize this.

Demographics (Age, Sex, Education):
Show some patterns but are weaker predictors than payment behavior.

Credit Limit & Age vs. Default



Clients who defaulted (1) tended to have lower credit limits than those who did not (0), suggesting credit limit is a useful sign for predicting default risk.

Feature Engineering

Four new summary features were created from existing columns to give the models a better picture of each client's repayment behavior:

Average Payment Status
How late was this client on average across all 6 months?

Total Bill Amount
What was the client's total outstanding debt across all months?

total payment Amounu
How much did the client actually pay back in total?

Payment Ratio
What fraction of their total bill did the client actually pay off? (A low ratio = more risk)

Models & Approach

Three models were trained on 70% of the data and tested on the remaining 30%:

Logistic Regression
Interpretable linear baseline model, works like a scoring system

Random Forest
Ensemble of decision trees that vote on the outcome; captures non-linear relationships.

LightGBM
Powerful radient boosting framework; fastest and most accurate.

class_weight='balanced'
applied to all models to address the 22% default class imbalance.

Model Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Log. Reg.	0.814	0.727	0.253	0.376	0.719
Rand. Forest	0.811	0.639	0.339	0.443	0.755
LightGBM ★	**0.760**	**0.467**	**0.594**	**0.523**	**0.767**

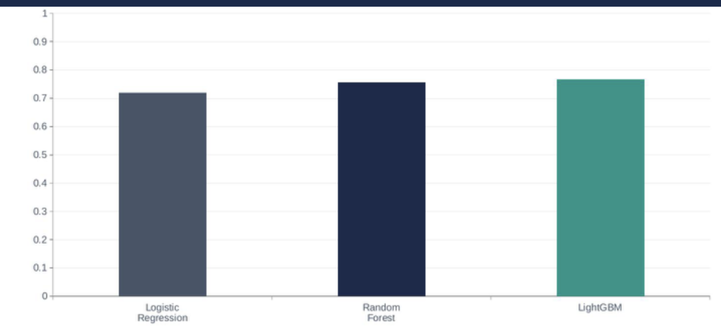
What do these metrics mean?

Accuracy:% of all predictions that were correct. High accuracy can be misleading with imbalanced data.

Recall: Of all actual defaulters, how many did the model catch? Higher recall = fewer missed defaulters.

ROC-AUC: Overall ability to distinguish defaulters from non-defaulters. Closer to 1.0 = better. LightGBM scored highest (0.767).

ROC-AUC by Model (Higher = Better)



Conclusions

LightGBM was the best model, achieving the highest ROC-AUC (0.767) and the best balance of catching defaulters without too many false alarms.